

Junyi (Conny) Zhou

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EDUCATION

- Harvard University** Boston, MA  
*M.S. in Health Data Science* August 2025 – December 2026
  - Coursework: Deep Learning, Healthcare Machine Learning
- Emory University, College of Arts and Sciences** Atlanta, GA  
*B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925* August 2021 – May 2025
  - Honors: Dean’s List Spring 2022, Spring 2023
  - Coursework: Machine Learning, NLP, Probability and Statistics, Data Structures, Database Systems

PROGRAMMING SKILLS

- Languages:** Python, C++, Java, SQL, R, JavaScript
- ML Frameworks:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
- Databases:** MySQL, PostgreSQL, MongoDB, Pinecone (Vector DB), Hadoop, Spark
- Cloud:** AWS (SageMaker, Lambda, EC2, S3, Redshift), Snowflake, BigQuery
- Tools:** Claude Code, Cursor, Git, Docker, Kubernetes, Jupyter
- Certifications:** AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

RESEARCH PROJECTS

- Airway Management Simulation Chatbot** Atlanta, GA  
*Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital* January 2024 – Present
  - Architected a clinical **agentic LLM pipeline** via a **RAG** system and **OpenAI models**, minimizing hallucinations and ensuring high data privacy for simulations.
  - Built a **HIPAA-compliant** full-stack platform (**Python/Flask**, React, MongoDB, AWS) tracking resident performance metrics via secure, real-time analytics.
  - Implemented **Prompt Engineering** and context window management to drive dynamic scenario generation for rapid cycle deliberate practice clinical training.
  - Led an Agile team of 6, facilitating code reviews and integrating clinician feedback to continually refine the NLP pipeline. <sup>1</sup>
- Reimplementation of Transformer Architecture** Atlanta, GA  
*Machine Learning Researcher — Emory Dept. of Computer Science* October 2024 – January 2025
  - Implemented a Transformer in **PyTorch** from scratch, building **Multi-Head Self-Attention**, **Positional Encodings**, and **Layer Normalization** to reproduce benchmarks.
  - Conducted extensive hyperparameter tuning and optimized the **Cross-Entropy Loss** function, achieving a 9.38 BLEU score on English-to-German translation.
  - Benchmarked custom implementation against standard `nn.Transformer` modules to strictly validate architectural correctness and training efficiency. <sup>2</sup>
- Cross-Species Drug Translation using Optimal Transport & Generative AI** Boston, MA  
*Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute* February 2026 – Present
  - Spearheaded computational frameworks using **Optimal Transport**, **VAEs**, and **diffusion models** to predict animal-to-human clinical trial translation via scRNA-seq atlases.
  - Engineered a custom **PyTorch** CellOT implementation from scratch, modeling complex species and treatment effects directly within VAE latent spaces.
  - Designed **scanpy** pipelines for automated cell-type labeling, integrating massive public and lab-generated scRNA-seq datasets for cancer and autoimmunity applications.
  - Trained out-of-distribution models mapping cellular subpopulations (e.g., regulatory vs. helper T cells) across species, evaluating GENOT to optimize T cell matching.
- AlphaFold Protein–Nucleic Acid Interaction Prediction** Atlanta, GA  
*Data Analyst — Bou-Nader Lab, Emory Dept. of Biochemistry* February 2025 – August 2025
  - Engineered a pipeline integrating **AlphaFold-Multimer** and **AlphaFold3** architectures to predict complex protein–nucleic acid interactions using **MSA** data.

<sup>1</sup><https://d3g1qw60hz7yw7.cloudfront.net>

<sup>2</sup><https://github.com/JunyiZhou-Conny/Reimplementation-of-Transformer>

- Automated high-throughput inference on **HPC** clusters with **parallel scheduling** to efficiently process structural predictions across GPU nodes.
- Developed **Python** scripts (PyMOL, ChimeraX) filtering predicted structures via **pLDDT** and **PAE** scores, selecting top-performing complexes for wet-lab validation.
- Authored CLI tools and technical documentation to democratize advanced ML inference tools for non-technical researchers.<sup>3</sup>

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<sup>3</sup><https://github.com/JunyiZhou-Conny/AlphaFold-Bou-Nader-Lab>